Nathan Abebe

+1 (203) 994-2951 | nathan.abebe@yale.edu | nathanabebe.com | linkedin.com/in/nathan5563 | github.com/Nathan5563

Education

Yale University New Haven, CT
B.S./M.S. Computer Science, B.S. Electrical Engineering | 3.87/4.00 May 2028

- Relevant coursework: Computer Architecture, Computer Networks, Computer Systems, Algorithms, Probability Theory, Linear Algebra, Discrete Math

Experience

Cloudflare May 2026 – Aug 2026
Software Engineer Intern Full-time, Austin, TX

- Integrated observability and data mining to optimize notification delivery for the enterprise SASE client, saving $3,000,000/mo.
- Rearchitected fan-out notification delivery by sharding 1,000,000 devices across a tree of serverless workers, with P99 and P99.99 latency improvements of 12 min to 50 sec and 11 hr to 3 min respectively.

Yale School of Engineering and Applied Science Aug 2025 – May 2026
Teaching Assistant | Computer Systems Part-time, New Haven, CT

- Supported 120+ undergraduate students by hosting weekly office hours and exam review sessions for Yale's hardest (student-voted) course.

Computer Graphics Group (Yale University) Aug 2025 – May 2026
Performance Engineering Researcher | Advisor: Dr. Mike Shah Part-time, New Haven, CT

- Implemented multithreaded Wavefront parser with SIMD, outperforming industry-standard tinyobjloader by 9x.
- Profiled and optimized critical paths (mmap IO, false sharing, zero-allocation) for a 500x end-to-end speedup over the initial single-threaded implementation.

Efficient Computing Lab (Yale University) May 2025 – Aug 2025
Systems Engineering Researcher | Advisor: Prof. Lin Zhong Full-time, New Haven, CT

- Engineered a virtual file system for multi-FPGA control, replacing per-experiment reconfiguration ($\approx$80% less engineering time); adopted by DECONET/HELIOS, a production distributed quantum error-correction decoder.
- Achieved sub-minute end-to-end remote FPGA configuration by building server-FPGA communication with a custom reliability layer over UDP (ACKs, retransmission, CRCs) guaranteeing delivery of control packets.

Projects

Quark – 2D Game Engine | *Dlang, ECS Architecture, SDL, GTK* Oct 2025 – Dec 2025

- Architected a 10k+ LoC parallelized 2D game engine with an entity-component system and event handlers.
- Implemented an interpreter for QScript: a specialized scripting language for use within the engine.
- Created applications on top of the engine, including a tilemap editor and various demo games.

ntraycer – Path Tracer | *Dlang, Path Tracing, Physics, Rendering* Mar 2025 – Jul 2025

- Engineered a Monte Carlo path tracer with zero external dependencies for generating photorealistic images.

nnes – NES Emulator | *Rust, Computer Architecture, Debugging* Mar 2025 – Jul 2025

- Built 5k+ LoC NES emulator in Rust, down to the various graphical glitches in the NES (plays Donkey Kong!).

TD-QAP – MIT iQuHack D-Wave 1st Place Prize | *Python, Quantum Annealing* Jan 2025

- Approximated the time-dependent quadratic assignment problem with D-Wave's quantum annealers.

Technical Skills

Languages: C, C++, Rust, Dlang, Python, Golang, TypeScript, Verilog, x86 Assembly
Tools & Libraries: Git, GDB, Valgrind, Vivado, Altium Designer, LTspice, SDL, GTK, OpenGL
Embedded: STM32, Linux, Yocto, bare-metal, PCB design, FPGA design
Protocols: HTTP, UDP, TCP, UART, SPI, I^{2}C